



SUMMER
OF MACHINE
LEARNING
AT SKOLTECH

Uncertainty in Diffusion and Flow Matching Models

Pre Defense report

Contributors: Gurgen Melikian,
Pavel Podlesnyi, Amir
Nigmatullin, Hajira Amjad
Curator: Nina Konovalova

[Project's repository](#)

Background & Motivation

Diffusion Transformer:

- Replaces U-Net backbone with a plain Vision Transformer on latent patches.
- Processed by every layer: $D_\theta(\mathbf{x}) = L_{B-1} \circ \dots \circ L_0$.
- Attention cost scales quadratically with token count $\rightarrow$ slow, expensive training.

TREAD

- Defines a route $\mathbf{r}_{i \rightarrow j}$: a random subset of tokens bypasses layers $L_i, \dots, L_j$ via identity, then rejoins the full sequence at layer $j+1$.

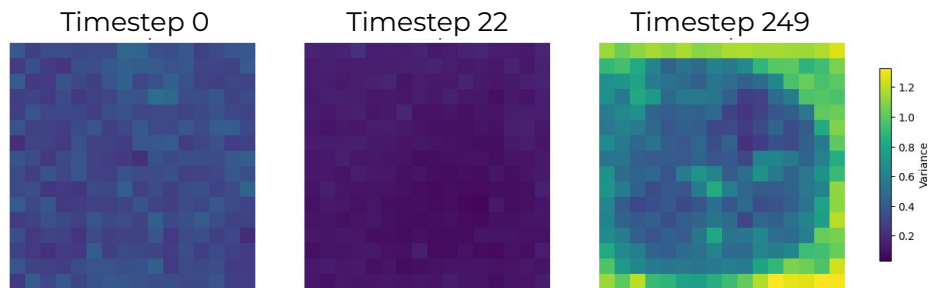
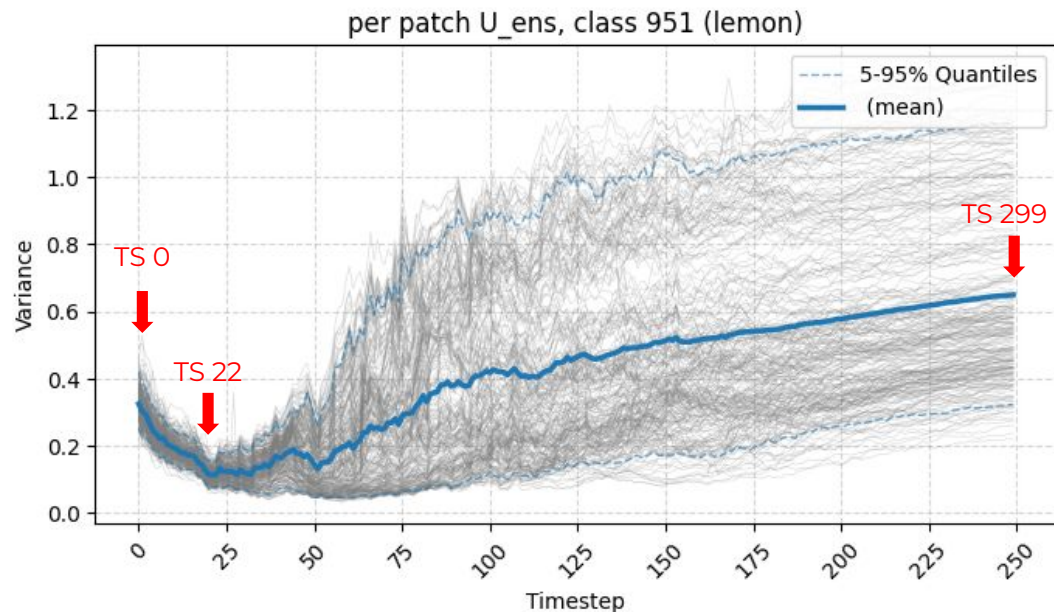
Project Goal & Objectives

Goal: build a training-free framework to quantify uncertainty in DiT and evaluate generation quality.

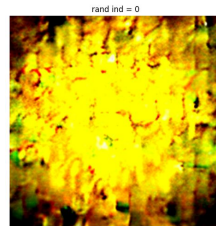
Main objectives:

1. Implement 3 uncertainty proxies (seed-ensemble/routing variance and attention entropy)
2. Profile uncertainty dynamics over timesteps and space (patch-level)
3. Link uncertainty to generation quality

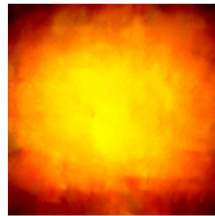
Seed-ensemble uncertainty



Timestep 0



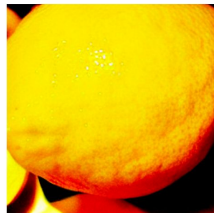
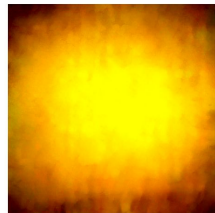
Timestep 22



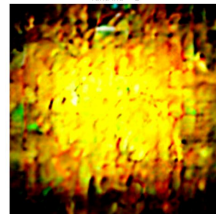
Timestep 249



rand ind = 1

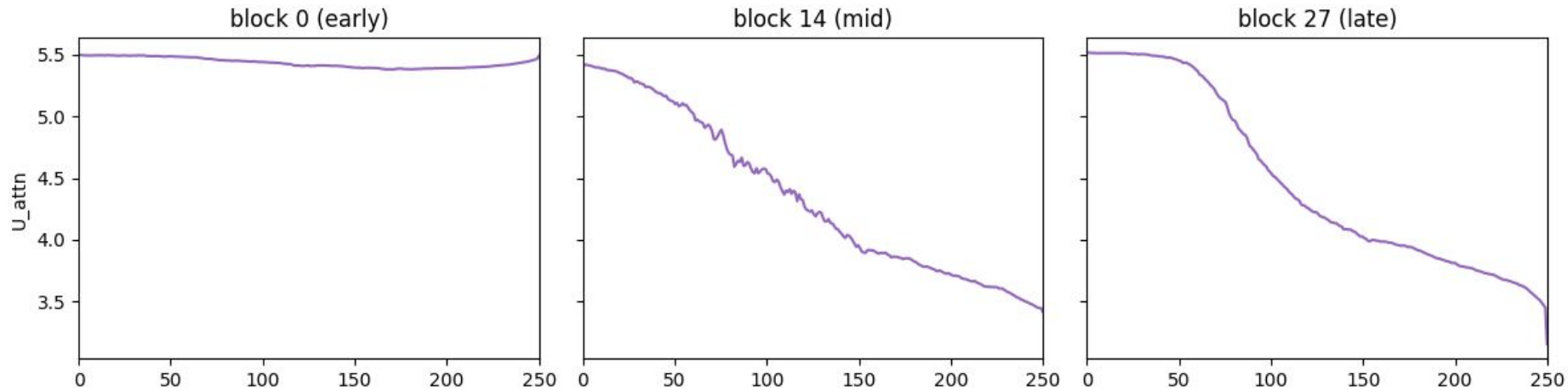


rand ind = 2



Attention entropy

Per-block attention entropy, class 951



Attention entropy across the denoising trajectory, shown separately for an early (block 0), mid (block 14), and late (block 27) DiT block (class 951).

Conclusion & Future Roadmap

Using uncertainty proxies reveal structure that existing training/sampling methods ignore.

- **Training:** use uncertainty profiles to guide *where* to route/skip tokens — pair proxies with TREAD-style routing, skipping low-entropy (already-confident) blocks/timesteps instead of routing uniformly at random.
- **Inference:** uncertainty-adaptive sampling — allocate more solver steps to high-entropy regions of the trajectory, fewer to low-entropy ones, as a training-free speedup
- Validate against fixed-rate baselines (TREAD's uniform 50%, standard fixed-step solvers) on FID.
- Extend profiling across more classes/seeds to confirm generalization.